# *Design & Implementation of Thermistor based Digital Thermometer*


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***Abstract*— An electrical device that shows the measured temperature in digital output is digital thermometer. Here, we have used a Negative Temperature Coefficient (NTC) thermistor at this regard. This type of thermistors is made from a semiconducting material that's been heated and compressed to form a temperature sensitive conducting material.**



***Keywords*— *NTC thermistor, Microcontroller, ATmega328P, Resistor, Capacitor, I2C LCD***


## I. Introduction

Analogue thermometers are mostly made of Mercury. Their readings are observation dependent. For very subtle measurements, these are not very accurate. We can't rely on analog thermometer where we need to measure very subtle temperature changes. For this purpose, digital thermometers are needed. We have made a digital thermometer with a NTC thermistor. Thermistors are rheostats that change their resistance with temperature change. For this purpose, we have used a 10k thermistor. A microcontroller was also included to convert resistance changes to temperature changes. To show the digital output, we included a I2C LCD as monitor.

## II. Methodology

This Project of building a digital thermometer based on Thermistor were implemented using a ATmega328P microchip. In this section we will discuss our design and implementation process at length.

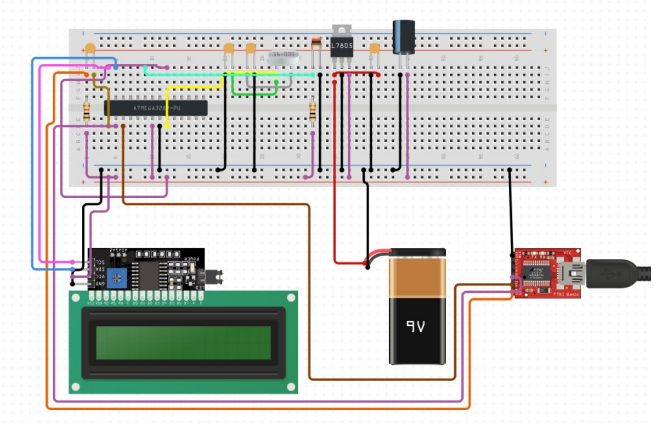
**Design & Component Selection:** we selected our components after much consideration. We used ATmega328P as our microchip, as it has a low power consumption and can execute 131 instructions per clock cycle. We used a 16MHz crystal oscillator to generate the clock cycle. To display the output in text format we used an I2C LCD. This is a 16x2 LCD soldered to an I2C adapter. As I2C has only 4 pins rather than 16 pins of 16x2 LCD, it makes the coding process easier. We used resistors to create a voltage divider circuit along with NTC thermistor.

Used components:

- ATmega328P microchip
- Breadboard
- 10K NTC thermistor
- 10K resistor
- 22 pF capacitor
- 16 MHz crystal oscillator
- Voltage Regulator (7805)
- I2C LCD
- 9V Battery
- Male to Male jumper cable

Circuit Diagram:

This circuit connection was designed in circuito.io app.




**Data Acquisition & Equation**: For converting voltage levels (due to changing resistance) to temperature, we need to calculate the general relation equation between temperature and voltage level. For this, we acquired data for different temperature & voltage levels. From these data we find our equation to convert voltage change to temperature.

Temperature = 0.2*Digital equivalent of voltage – 82 (1)

## III. Operating Principle

Here we used a Voltage divider circuit as primary input connected to pin 24 (A1) of our microcontroller. This divider circuits output value changes with change of resistive value of NTC thermistor. For voltage dividing circuit we used 10KΩ resistor and 10KΩ thermistor in parallel. To display our output data, we connected SCL & SDA of I2C adapter to pin 28 (A5) & 27 (A4) respectively. VCC & GND of I2C adapter were connected to 5V supply and ground respectively. Voltage changes inputted through A1 is converted according to (1). This value is then sent to A4 & A5, and the output is finally shown in the LCD.

## IV. Cost Analysis

| Component | Price (BDT) |
|---|---|
| ATmega328P microchip | 550 |
| Breadboard | 100 |
| 10K NTC thermistor | 10 |
| 10K resistor | 5 |
| 22 pF capacitor | 5 |
| 16 MHz crystal oscillator | 12 |
| Voltage Regulator (7805) | 10 |
| I2C LCD | 350 |
| 9V Battery | 50 |
| 9V Battery connector | 10 |
| Male to Male jumper cable | 50 |
| Total | 1152 BDT |

## V. Output




## VI. Applications

1. Digital thermometers are used to monitor room temperature.
2. Digital thermometers are used to give temperature reading of air, atmosphere & water.
3. Also used for home healthcare, cooking, or as temperature indicator of refrigerator.
4. Used in Laboratory for monitoring experiments & chemical reaction temperature.

## VII. Future Scope

We can use digital thermometer in various field of measuring temperature. The digital thermometer can be used in cold storage. Further modification of This project can be made so that if the temperature value crosses a specific value, an alarm will ring. In same manner, it could be used as fire alarming system, which we can set up in chemical storage, and other storage places where a threshold temperature needs to be maintained.

## VIII. Conclusion

We have tried different designs of thermistor based digital thermometer. But we decided to use this configuration of thermistor paired with ATmega328P microchip. This project can be modified to include control systems which will increase its field of application.


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